

Denominator-Adjusted Analysis of COVID-19 VAERS Reporting Patterns

Using Public VAERS and CDC Lot Distribution Data

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This report presents a denominator-adjusted analysis of publicly available VAERS and CDC lot-distribution data. The findings identify reporting patterns for further investigation and do not establish causation, clinical incidence, or product defects.

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Executive Summary

This report examines COVID-19 vaccine adverse-event reports submitted to the Vaccine Adverse Event Reporting System (VAERS) and combines those records with publicly released CDC lot-distribution data to evaluate reporting patterns at the manufacturer and vaccine-lot level.

The central question was narrow:

Do apparent lot-level reporting patterns remain visible after accounting for distribution volume?

That question matters because raw VAERS report counts alone are incomplete. A vaccine lot associated with a large number of reports may simply have been a large lot distributed broadly across the country. Without denominator information, meaning some measure of how many doses were distributed or administered, lot-level comparisons are limited.

This project incorporated publicly released CDC lot-distribution records, allowing calculation of denominator-adjusted reporting metrics for matched Pfizer lots. These included reports per million doses shipped, serious reports per million doses shipped, death reports per million doses shipped, expected serious reports, observed-versus-expected serious-reporting ratios, and approximate confidence intervals.

Several apparent lot-level signals diminished after denominator adjustment, indicating that distribution volume explained some raw-count differences.

However, several lots remained elevated after denominator adjustment. The largest denominator-adjusted serious-reporting elevation observed in this analysis was Pfizer lot GJ3277, which showed an observed-versus-expected serious-reporting ratio of 5.049 with an approximate 95% confidence interval of 4.638 to 5.460. The EN6200, EN6201, and EN6202 lots also remained elevated, with observed-versus-expected ratios of 2.719, 2.855, and 2.693, respectively.

The primary conclusion supported by this analysis is this:

Within the matched Pfizer lots analyzed, adjustment for shipped-dose volume reduced some apparent signals while others remained elevated.

These findings do not establish causation, clinical incidence, manufacturing defects, or product-quality differences.

They do show that shipped-dose adjustment changed the interpretation of lot-level VAERS reporting patterns and identified several lots as reasonable candidates for follow-up analysis using stronger independent datasets.

Introduction

Since 2021, debates surrounding COVID-19 vaccine safety have frequently referenced VAERS. Those discussions have often produced more disagreement about the data than agreement.

Supporters of the vaccination campaign have frequently argued that VAERS cannot be used to draw meaningful conclusions because it is a passive reporting system that does not establish causation. Critics have pointed to the volume and nature of reported events as evidence that important safety signals were overlooked or dismissed.

One criticism appeared repeatedly throughout these debates and, in my view, deserved serious consideration: lot-level analyses based solely on raw VAERS report counts lack denominator data. A vaccine lot associated with a large number of reports may simply reflect a large number of distributed doses. Without information

regarding distribution volume, it is difficult to determine whether a particular lot generated an unusual reporting frequency or merely represented a larger share of the vaccination campaign.

This project began with a straightforward question:

Do apparent lot-level reporting patterns remain visible after accounting for distribution volume?

Answering that question required combining publicly available VAERS data with publicly released lot-distribution records and constructing a denominator-adjusted analytical framework.

I approached this project with substantial skepticism regarding the transparency of COVID-era public-health decision making and vaccine-safety monitoring. That skepticism was one reason for undertaking the analysis. At the same time, skepticism is most valuable when applied consistently. For that reason, the objective was not to confirm a preexisting conclusion but to determine whether observed reporting patterns persisted when more complete data became available.

Some apparent signals weakened after denominator adjustment.

Others did not.

The results presented in this report reflect the outcome of that process.

Researcher Disclosure

I approached this project with substantial skepticism regarding the transparency of COVID-era public-health decision making, vaccine-safety monitoring, and the adequacy of public access to relevant safety data.

That skepticism was one reason for undertaking the analysis.

At the same time, skepticism must be applied consistently. A serious analysis cannot assume that every apparent signal is meaningful simply because it appears to support a skeptical position. Nor can it dismiss every signal merely because VAERS has known limitations.

For that reason, the project was designed to test whether apparent reporting signals remained after the introduction of denominator data rather than assuming that initial observations were correct.

Several signals diminished after denominator adjustment. Several others persisted.

The findings presented in this report reflect the results of the analysis rather than any conclusion that existed before the analysis began.

Objectives

The objectives of this project were:

1. Evaluate manufacturer-level reporting patterns within COVID-19 VAERS data.
2. Evaluate lot-level reporting patterns.
3. Incorporate publicly released CDC lot-distribution data where available.
4. Calculate denominator-adjusted reporting metrics.
5. Calculate observed-versus-expected serious-reporting ratios.
6. Estimate approximate confidence intervals for elevated lot-level findings.
7. Produce a transparent and reproducible analytical framework using publicly available data.

Data Sources

The analysis used the following data sources:

- Public VAERS report data
- Public VAERS vaccine data
- Public VAERS symptom data
- Publicly released CDC lot-distribution records

- Manufacturer fields
- Vaccine lot identifiers
- Serious adverse-event indicators
- Death indicators
- Hospitalization, life-threatening, and disability indicators
- Symptom records

Data processing and analysis were performed using DuckDB, Python, Pandas, and related statistical libraries. The repository, scripts, output files, and documentation were publicly released to support independent review.

Data Availability and Access Challenges

One of the practical obstacles encountered during this project involved access to lot-level distribution information.

During the course of the research, I attempted to obtain vaccine lot information through the CDC's COVID-19 Vaccine Lot Number and Expiration Date system. Access to that system was denied.

The denial itself is less important than its consequence. Without denominator information, independent researchers are limited to evaluating raw adverse-event reports. Meaningful lot-level analysis requires both a numerator, reported events, and a denominator, distribution volume or doses administered.

For several years, this limitation represented one of the strongest criticisms of lot-based VAERS investigations. In many cases, that criticism was justified.

The denominator-adjusted portion of this project became possible only after incorporating publicly released lot-distribution records that became available through separate disclosure efforts.

Whether one ultimately agrees with the conclusions of this report or not, the experience underscores the importance of transparency. Independent review is only possible when independent researchers have access to the information necessary to evaluate competing claims.

Methodology

Phase 1: Baseline VAERS Analysis

The first phase of the project analyzed COVID-related VAERS reports by manufacturer, lot number, state, age group, and symptom grouping.

Initial analyses examined:

- Total reports by manufacturer
- Serious reports by manufacturer
- Death reports by manufacturer
- Serious-reporting percentages
- Lot-level report counts
- Age-bucket patterns
- State-level reporting patterns
- Symptom clusters

Targeted event categories, including myocarditis, pericarditis, thrombosis, pulmonary embolism, deep vein thrombosis, cerebral venous sinus thrombosis, Guillain-Barré syndrome, syncope, and loss of consciousness

These analyses identified manufacturer-level differences and several lots that repeatedly appeared in elevated-reporting analyses.

Phase 2: Denominator Adjustment

The second phase incorporated publicly released CDC lot-distribution records. The purpose was to address the denominator problem in lot-level VAERS analysis.

For matched Pfizer lots, the following metrics were calculated:

- Total VAERS reports
- Serious reports
- Death reports
- Total doses shipped
- Reports per million doses shipped
- Serious reports per million doses shipped
- Deaths per million doses shipped
- Expected serious reports
- Observed-versus-expected serious-reporting ratio
- Approximate 95% confidence interval

Expected serious reports were calculated using the overall serious-reporting baseline among matched Pfizer lots with denominator information.

The denominator used in this analysis was doses shipped, not confirmed doses administered. That distinction is important. Doses shipped provide a substantially better denominator than raw report counts alone, but they are not equivalent to confirmed administered doses.

Study Purpose and Interpretation

This analysis should be viewed as a hypothesis-generating signal-detection study rather than a causal safety study. The objective is to identify patterns warranting additional investigation, not to establish causation.

The findings are intended to support further review, replication, and evaluation using additional data sources rather than serve as definitive evidence regarding vaccine safety, product quality, or clinical risk.

Understanding Observed-Versus-Expected Ratios

The observed-versus-expected ratio, abbreviated O/E, compares the number of serious reports actually observed for a lot with the number expected based on the overall baseline.

An O/E ratio of 1.0 means the lot matched expectation.

An O/E ratio greater than 1.0 means the lot had more serious reports than expected.

An O/E ratio below 1.0 means the lot had fewer serious reports than expected.

For example, an O/E ratio of 2.0 means approximately twice as many serious reports were observed as expected under the baseline used in this analysis.

An O/E ratio of 5.0 means approximately five times as many serious reports were observed as expected.

Confidence intervals were calculated to estimate statistical uncertainty. When a confidence interval remains entirely above 1.0, the elevation is less likely to be explained by random variation alone within the framework of this analysis.

Relationship to Existing Safety Literature

The findings in this report should be interpreted within the broader COVID-19 vaccine safety-monitoring literature.

VAERS is a passive surveillance system and does not establish causation by itself. However, passive surveillance systems are used for signal detection, and several COVID-19 vaccine safety signals initially

identified or monitored through post-market surveillance were later discussed in official safety communications and peer-reviewed publications.

The myocarditis and pericarditis findings in this project are consistent with broader literature identifying myocarditis as a rare but serious adverse event reported after mRNA COVID-19 vaccination, particularly among younger males.

The Janssen-associated thrombosis and Guillain-Barré findings are also consistent with prior CDC, FDA, and peer-reviewed safety discussions identifying thrombosis with thrombocytopenia syndrome and Guillain-Barré syndrome as rare but serious safety concerns following Janssen COVID-19 vaccination.

This report does not rely on those publications to prove the lot-level findings. Rather, those publications provide context showing that several event categories identified in this analysis have also been recognized in official or peer-reviewed safety-monitoring literature.

The primary contribution of this report is narrower: it evaluates whether certain lot-level VAERS reporting patterns remain elevated after incorporating lot-distribution denominator data.

Findings

Manufacturer-Level Findings

The baseline analysis identified persistent reporting-pattern differences among manufacturers.

At the manufacturer level, earlier phases of the analysis found that Pfizer-associated reports were more enriched for myocarditis and pericarditis terms, while Janssen-associated reports were more enriched for thrombosis-related terms, cerebral venous sinus thrombosis, pulmonary embolism, deep vein thrombosis, Guillain-Barré syndrome, syncope, and loss of consciousness.

These manufacturer-level findings were broadly consistent with known safety-monitoring discussions in the medical and regulatory literature. The manufacturer-level analysis was not designed to estimate clinical incidence rates. It was designed to identify reporting-pattern differences within VAERS.

Lot-Level Findings After Denominator Adjustment

The denominator-adjusted analysis produced the central lot-level findings in this project.

Several Pfizer lots remained elevated after accounting for doses shipped.

Lot	Doses Shipped	Serious Reports	Expected Serious Reports	O/E Ratio	Approx. 95% CI
GJ3277	3,930,955	579	114.68	5.049	4.638 to 5.460
EN6201	15,043,577	1,253	438.89	2.855	2.697 to 3.013
EN6200	13,890,331	1,102	405.25	2.719	2.559 to 2.880
EN6202	13,619,726	1,070	397.35	2.693	2.531 to 2.854
ER2613	15,392,071	926	449.06	2.062	1.929 to 2.195

The largest denominator-adjusted serious-reporting elevation observed in this analysis was lot GJ3277. It had 579 serious reports compared with 114.68 expected serious reports, producing an O/E ratio of 5.049. The confidence interval remained well above 1.0.

The EN6200, EN6201, and EN6202 lots also remained elevated after denominator adjustment. These lots had large distribution volumes, ranging from approximately 13.6 million to 15.0 million doses shipped, yet their serious-reporting frequencies remained elevated relative to the Pfizer baseline used in this analysis.

This finding is important because shipped-dose adjustment did not eliminate the observed reporting differences.

Signals That Diminished After Denominator Adjustment

Several lots that appeared notable when examined through raw report counts became less remarkable after denominator adjustment.

This result is important for two reasons.

First, it confirms that denominator adjustment matters. Some apparent lot-level signals were at least partly explained by lot size.

Second, it shows that the analysis did not merely preserve every initially observed signal. Some findings weakened when better denominator information was introduced.

That makes the lots that remained elevated appropriate candidates for follow-up analysis, while still requiring replication and confounder assessment.

Understanding the Findings in Plain English

Many readers are not statisticians, epidemiologists, or data scientists. This section explains the core finding in plain language.

Imagine two vaccine lots.

Lot A generated 1,000 adverse-event reports.

Lot B generated 500 adverse-event reports.

At first glance, Lot A appears worse.

But what if Lot A had 20 million doses distributed, while Lot B had only 2 million?

Once distribution volume is considered, Lot B may actually have generated more reports relative to its size.

That is the denominator problem.

One of the primary goals of this project was to address that problem. Raw report counts can be misleading because they do not show how many doses were distributed from each lot.

After denominator adjustment, some apparent signals disappeared or became less remarkable. That means those lots may have looked unusual mainly because they were large or widely distributed.

Other signals remained elevated even after accounting for distribution volume.

In plain English:

Some lot-level concerns became weaker when better data were added.

Others remained.

Those remaining reporting patterns are reasonable candidates for further investigation rather than dismissal based solely on lot size.

What the Analysis Can Reasonably Tell Us

The findings support several conclusions.

- First, meaningful lot-level reporting differences existed within the analyzed data.
- Second, some differences persisted after denominator adjustment.
- Third, adjustment for shipped-dose volume did not eliminate all observed lot-level reporting differences.
- Fourth, access to denominator data materially improves the quality of lot-level analysis.

The findings do not establish biological causation, manufacturing defects, product contamination, formulation differences, clinical incidence rates, or individual injury determinations.

However, the results show that adjustment for shipped-dose volume changed the interpretation of lot-level VAERS reporting patterns and did not eliminate all observed reporting differences within the analyzed data. That is the primary conclusion supported by this analysis.

Underreporting Considerations

VAERS is a passive reporting system. Passive reporting systems are widely understood to capture only a portion of adverse events. The precise degree of underreporting varies by event type, severity, public awareness, reporting practices, and other factors.

This matters for interpretation.

If adverse events are underreported, then the absolute number of events reflected in VAERS may be lower than the true number of events that occurred.

However, underreporting alone does not automatically explain why certain lots exhibit substantially higher reporting frequencies than other lots. If underreporting were uniform across all lots, it would affect absolute counts but would not necessarily erase relative differences. If underreporting varied by geography, population, time period, provider behavior, or public awareness, it could influence lot-level comparisons.

For that reason, underreporting remains an important limitation, but it does not remove the need to investigate denominator-adjusted lot-level differences that persist within the available data.

Regulatory and Safety Context

This report does not independently evaluate FDA authorization decisions, product approval standards, manufacturing quality controls, or clinical trial evidence. The analysis is limited to publicly available adverse-event reports and publicly released lot-distribution records.

However, the findings are relevant to post-authorization safety monitoring and transparency.

COVID-19 vaccines were first distributed under emergency-use authorization frameworks that relied on FDA-issued fact sheets and evolving safety communications. Later, certain products received formal approval and prescribing information. Safety labeling also continued to evolve after rollout, including later FDA-required updates related to myocarditis and pericarditis warnings for mRNA COVID-19 vaccines.

In that context, denominator-adjusted lot-level reporting differences should not be dismissed merely because VAERS cannot prove causation by itself. Passive surveillance systems are not designed to provide final answers. They are designed to detect signals that may require further investigation.

The central regulatory question is therefore not whether VAERS alone proves causation. It does not. The question is whether elevated reporting patterns, especially those that persist after denominator adjustment, should trigger deeper review, public disclosure, and independent investigation.

At minimum, the findings strengthen the case for greater transparency regarding lot-level distribution data, active-surveillance findings, manufacturing records, quality-control records, and internal safety analyses.

When products are distributed to millions of people under extraordinary public pressure and emergency-policy conditions, agencies and manufacturers carry a heightened ethical responsibility to disclose relevant safety information clearly, promptly, and completely.

Ethical and Transparency Considerations

This report does not attempt to determine whether regulatory decisions were correct, whether authorization standards were appropriate, or whether any individual adverse event was caused by vaccination.

However, the findings do highlight the importance of transparency in public-health decision making.

The ability to independently evaluate safety signals depends upon access to relevant data. Meaningful lot-level analysis requires more than adverse-event reports alone. It requires denominator information, methodological transparency, and the ability of independent researchers to reproduce or challenge findings.

The experience of attempting to obtain lot-level information during this project illustrates the practical consequences of limited data access. Independent review is weakened when researchers cannot obtain the information necessary to evaluate competing claims.

Public confidence is strengthened not by restricting scrutiny but by enabling it. Transparency allows conclusions to be challenged, corrected, replicated, or confirmed. In matters affecting millions of people, that transparency is not merely a scientific virtue. It is an ethical responsibility. Where meaningful safety signals persist, public confidence is best served by investigation and disclosure rather than dismissal.

Limitations

This analysis has several important limitations.

VAERS Does Not Establish Causation

VAERS is a passive surveillance system. A report submitted to VAERS does not prove that a vaccine caused the reported event.

Doses Shipped Are Not Doses Administered

The denominator analysis used doses shipped rather than confirmed doses administered. Doses shipped provide a useful distribution denominator, but they are not equivalent to administered-dose records.

Reporting Frequency Is Not Incidence

The rates calculated in this report are reporting rates, not clinical incidence rates. They should not be interpreted as true population risk.

Lot Data Required Cleaning and Normalization

Lot identifiers in public adverse-event datasets are often inconsistent. Lot-number fields may contain typographical errors, formatting issues, missing values, or malformed entries. Normalization was necessary to match records across datasets.

Residual Confounding May Remain

Lot-level reporting patterns may be influenced by factors not fully accounted for in this analysis, including geography, age distribution, reporting behavior, time period, public awareness, provider reporting practices, and distribution channel.

The Analysis Is Limited to Available Public Data

The analysis did not include internal manufacturer records, lot-testing records, complete administered-dose data, active-surveillance datasets, or clinical case review records.

Multiple Comparisons

A large number of vaccine lots were evaluated during the course of this analysis. When many lots are examined simultaneously, some extreme observations may occur by chance alone.

This analysis did not apply a formal multiple-comparison correction or false-discovery-rate adjustment. Consequently, elevated lots identified in this report should be viewed as candidates for further investigation and replication rather than definitive findings.

Future work should evaluate whether the observed signals persist under alternative statistical approaches and across independent datasets.

Future Research Priorities

This project should be viewed as a starting point rather than a final answer.

Future work could be strengthened through access to:

- Confirmed doses administered by lot
- State immunization registry data
- V-Safe data
- Vaccine Safety Datalink data
- FDA BEST or Sentinel data
- CMS and Medicare claims data
- Private insurance claims data
- Manufacturing and quality-control records
- Lot-specific testing records
- Clinical follow-up investigation records

Confirmed administered-dose denominators by lot remain one of the most important unresolved data elements. Access to administered-dose records would allow future analyses to evaluate whether observed denominator-adjusted patterns persist when distribution volume is replaced with actual administration volume.

Replication across independent datasets represents one of the strongest methods available for determining whether observed signals reflect meaningful safety patterns, reporting artifacts, or other explanations.

Conclusion

The central question of this project was not whether VAERS proves causation.

It does not.

The question was whether denominator-adjusted analysis changes the interpretation of apparent lot-level reporting patterns.

The answer is yes.

Several lots that appeared notable when evaluated using raw report counts became less remarkable after accounting for distribution volume. At the same time, several lots remained elevated after denominator adjustment, including GJ3277 and the EN6200, EN6201, and EN6202 cluster.

Those findings do not establish manufacturing defects, contamination, biological mechanisms, or clinical incidence rates. They do, however, show that adjustment for shipped-dose volume did not eliminate all observed reporting differences within the analyzed data.

Reasonable people may disagree about the implications of those findings. They should not have to disagree about the importance of transparency, reproducibility, and access to data.

The purpose of this report is not to provide the final word on vaccine safety. It is to document a reproducible analysis, identify patterns that persisted after denominator adjustment, and make the methodology available for independent review, criticism, replication, and improvement.

Several findings in this report would warrant re-evaluation if future analyses using administered-dose denominators, active-surveillance systems, claims databases, immunization registry data, or other independent data sources fail to replicate the observed patterns.

Replication across independent datasets remains one of the strongest methods available for distinguishing meaningful safety signals from statistical artifacts, reporting biases, or limitations inherent in passive-surveillance systems.

Scientific questions are best resolved through examination rather than assumption.

For that reason, all underlying methods, outputs, and documentation associated with this project have been released publicly.

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Appendix A – Lot Number Normalization and Matching Methodology

Public VAERS lot identifiers contain substantial variability and data-quality issues. To improve matching between VAERS reports and publicly released lot-distribution records, the following normalization procedures were applied:

1. Conversion of all lot identifiers to uppercase.
2. Removal of leading and trailing whitespace.
3. Removal of non-printable characters.
4. Standardization of common formatting inconsistencies where appropriate.
5. Exclusion of blank, missing, or obviously invalid lot entries.
6. Preservation of original identifiers where ambiguity existed.
7. Matching against denominator datasets using normalized lot identifiers.

No attempt was made to infer unknown lot numbers or reconstruct incomplete identifiers beyond basic normalization procedures.

All matching rules, scripts, and outputs are included within the public repository to support independent review and replication.